

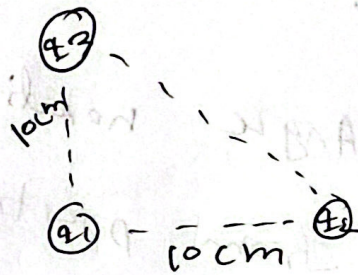
Assignment - 1

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section: 8

1. $q_1 = -5 \text{ nC}$
 $q_2 = 10 \text{ nC}$
 $q_3 = 20 \text{ nC}$



Force on q_2 due to $q_1 = F_{12}$

$$F_{12} = k \cdot \frac{|q_1 q_2|}{r_{12}^2} = \frac{(8.99 \times 10^9) (5 \times 10^{-9} \times 10 \times 10^{-9})}{(0.10)^2}$$

$$= 4.495 \times 10^{-5} \text{ N}$$

$$= 44.95 \mu\text{N}$$

$\vec{F}_{12} = -44.95 \mu\text{N} \hat{i}$ (negative x-direction)

Force on q_2 due to q_3

$$F_{32} = k \cdot \frac{|q_3 q_2|}{r_{32}^2} = \frac{8.99 \times 10^9 \times 20 \times 10^{-9} \times 10 \times 10^{-9}}{(0.10)^2}$$

$$= 179.8 \mu\text{N}$$

$\vec{F}_{32} = -179.8 \mu\text{N} \hat{j}$ (negative y-direction)

$F_{\text{net}} = F_{12} + F_{32} = -44.95 \mu\text{N} \hat{i} - 179.8 \mu\text{N} \hat{j}$

$|F_{\text{net}}| = \sqrt{(44.95)^2 + (179.8)^2} = 185.4 \mu\text{N}$

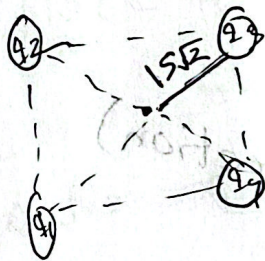
$\theta = \tan^{-1}(179.8/44.95) = \tan^{-1}(4.0) = 76.0^\circ$

$$F_{\text{net}} = (-44.999 - 179.85) \mu\text{N}$$

Magnitude Angle notation $|F_{\text{net}}| = 189.4 \mu\text{N}$
at 296.0° from positive x-axis

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$$q_1 = q_2 = q_4 = 10 \text{ nC}, \quad q_3 = -19 \text{ nC}$$



$$E_1 = E_2 = E_4 = k|q|/r^2$$

$$= 8.99 \times 10^9 (10 \times 10^{-9})$$

$$\frac{(0.1061)^2}{= 7.99 \times 10^3 \text{ N/C}}$$

$$E_3 = k|q_3|/r^2$$

$$= \frac{(8.99 \times 10^9) (19 \times 10^{-9})}{(0.1061)^2} = 1.198 \times 10^4 \text{ N/C}$$

$$E_{\text{net}} = 7.99 \times 10^3 + 1.198 \times 10^4$$

$$= 1.998 \times 10^4 \text{ N/C}$$

Direction Toward q_3

3)

$$q_1 = 9 \text{ nC}$$

$$q_2 = -30 \text{ nC}$$

Distance between charges = 10 cm

set up coordinate system.

• place q_1 at origin $(0,0)$

• place q_2 at $(0.10 \text{ m}, 0)$

Look for zero field point at distance x from q_1

$$E_1 = kq_1/x^2$$

$$E_2 = k|q_2|/(0.10 - x)^2$$

$$E_1 = kq_1/x^2$$

$$E_2 = k|q_2|/(0.10 + x)^2$$

$$kq_1/x^2 = k|q_2|(x - 0.10)^2$$

$$9/x^2 = 30(x - 0.10)^2$$

$$\Rightarrow 29x^2 + x - 0.09 = 0$$

$$x = 0.029 \text{ m} = 2.9 \text{ cm}$$

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rod length = 0.49m bent into circular ring

Total charge $Q = 3.2 \mu C$

$$\text{Radius, } r = L/2\pi = 0.49/2\pi = 0.0716m$$

a) Electric field 8cm from center

$$E = kQz / (z^2 + r^2)^{3/2}$$

$$k = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$Q = 3.2 \times 10^{-6} \text{ C}$$

$$z = 0.08m \quad r = 0.0716m$$

$$E = \frac{(8.99 \times 10^9)(3.2 \times 10^{-6})(0.08)}{[(0.08)^2 + (0.0716)^2]^{3/2}}$$

$$E = 1.87 \times 10^6 \text{ N/C}$$

5) Distance for maximum electric field

$$E = kQ \frac{z}{(z^2 + r^2)^{3/2}}$$

$$\frac{dE}{dz} = kQ \left[(z^2 + r^2)^{-3/2} - z \left(\frac{1}{2} \right) (z^2 + r^2)^{-5/2} \cdot 2z \right]$$

setting $\frac{dE}{dz} = 0$

$$(z^2 + r^2)^{-3/2} = z^2 (z^2 + r^2)^{-5/2}$$

$$z^2 + r^2 = 3z^2$$

$$z = \frac{r}{\sqrt{2}}$$

$$z = 0.0716 / \sqrt{2} = 0.0506 \text{ m} = 5.06 \text{ cm}$$